

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-12

Subject: Barometric condensers, steam ejectors and steam-jet vacuum units.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Brief description of the item: $P = 0.013$ atm (ejectors) and $P = 0.075$ atm (condensers); the steam-jet vacuum unit comprises an ejector, a condenser and a vacuum pump. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer's works) and DDP project site in north-west Russia terms (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from order / advance payment);
3. overall dimensions and weights of the equipment and of the largest shipping blocks (for logistics studies);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical requirements

To RFQ No. OVE75-RFQ-12 — values follow the Customer's input data and terms of reference; Basic Engineering (FEED) stage, subject to refinement at the next project stage.

RFQ item	Barometric condensers, steam ejectors and steam-jet vacuum units
Summary	$P = 0.013$ atm (ejectors) and $P = 0.075$ atm (condensers); the steam-jet vacuum unit comprises an ejector, a condenser and a vacuum pump

Specific requirements	The scope of supply of the vacuum system shall cover two independent crystallization lines
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Equipment covered by this request:

1. **Steam ejector** Copper sulphate crystallization · item 4.1.2

Duty	Creation of deep vacuum at stage II
Key parameters	288 kg/h of secondary vapour; design flow $Q = 375$ kg/h; $P = 0.013$ atm (10 mmHg)
Quantity	1
Materials	904L steel or equivalent
Dimensions, hold-up	Margin factor 1.3

2. **Steam ejector** Mixed sulphate crystallization · item 4.1.24

Duty	Creation of vacuum at stage II
Key parameters	69 kg/h of secondary vapour; design flow $Q = 90$ kg/h; $P = 0.013$ atm
Quantity	1
Dimensions, hold-up	Margin factor 1.3

3. **Barometric condenser** Copper sulphate crystallization · item 4.1.3

Duty	Condensation of the secondary vapour from the evaporators and of the ejector steam, irrigation with cold recirculating water
Key parameters	Flow 3.24 t/h; design flow $Q = 4.5$ t/h; $P = 0.075$ atm (55 mmHg)
Quantity	1
Dimensions, hold-up	Margin factor 1.4

4. **Barometric condenser** Mixed sulphate crystallization · item 4.1.25

Duty	Condensation of the secondary vapour, irrigation with cold recirculating water
Key parameters	Flow 9.1 t/h; design flow $Q = 12.0$ t/h; $P = 0.075$ atm
Quantity	1
Dimensions, hold-up	Margin factor 1.3

5. **Steam-jet vacuum unit (SJVU)** Crystallization · item 6, Fig. 4.4/4.5

Type / model	Ejector + SJVU condenser + vacuum pump
Duty	Maintaining the vacuum in the condenser
Quantity	1 per line

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

预算性技术商务报价询价函 编号 OVE75-RFQ-12

事由：大气冷凝器、蒸汽喷射器及蒸汽喷射真空机组。

KVANT LLC（俄罗斯联邦）正在开展铜冶炼厂建设项目的基础设计（FEED）——“焙烧—浸出—电积”联合工艺，年产阴极铜 75,000 吨——现请贵司就上述设备提交预算性技术商务报价（预算报价）。项目场地：俄罗斯西北部一处 在役冶金厂区。

设备简要说明：P = 0.013 atm（喷射器）及 P = 0.075 atm（冷凝器）；蒸汽喷射真空机组由喷射器、冷凝器和真空泵组成。技术要求见本询价函附件 1。

请在报价中包含：

1. 预算价格——分别按 EXW（制造商工厂）及 DDP 至俄罗斯西北部项目场地（Incoterms 2020）条件报价，注明币种及报价有效期；
2. 制造周期及预计交货期（自订单/预付款之日起按周计）；
3. 设备单机及最大运输单元的外形尺寸和重量（用于物流研究）；
4. 类似供货业绩表：产品、工况介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质，以及供货范围（仪表、备件、技术文件）；
6. 公用工程消耗及安装要求（如有资料）。

本询价为基础设计（FEED）阶段的预算性询价：不构成要约，双方均不因此承担义务；详细技术数据表将在项目下一阶段发出。

回复期限：请在收到本询价函后 3（三）周内将报价发送至 stepan@kvantpro.com。请确认收到本询价函。

保密条款。本询价函及随附技术资料仅供编制报价使用，属保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发布。提交报价不为双方设立任何法律义务。

联系人：Stepan Kharkov，KVANT LLC，stepan@kvantpro.com。

附件 1：本询价设备的技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-12 — 参数依据业主原始数据及设计任务书；基础设计（FEED）阶段，后续阶段可能进一步细化。

询价设备	大气冷凝器、蒸汽喷射器及蒸汽喷射真空机组
简要说明	P = 0.013 atm（喷射器）及 P = 0.075 atm（冷凝器）；蒸汽喷射真空机组由喷射器、冷凝器和真空泵组成
特殊要求	真空系统的供货范围须满足两条独立结晶生产线的要求

本询价涵盖的设备清单：

1. 蒸汽喷射器 硫酸铜结晶 · 位号 4.1.2

用途	在第二级产生深度真空
主要参数	二次蒸汽 288 kg/h；计算量 Q = 375 kg/h；P = 0.013 atm（10 毫米汞柱）
数量	1
材质	904L 钢或同等材质

尺寸、滞留量	裕量系数 1.3
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2. 蒸汽喷射器 混合硫酸盐结晶 · 位号 4.1.24

用途	在第二级产生真空
主要参数	二次蒸汽 69 kg/h ; 计算量 $Q = 90 \text{ kg/h}$; $P = 0.013 \text{ atm}$
数量	1
尺寸、滞留量	裕量系数 1.3

3. 大气冷凝器 硫酸铜结晶 · 位号 4.1.3

用途	冷凝蒸发器的二次蒸汽及喷射器蒸汽，采用冷却循环水喷淋
主要参数	流量 3.24 t/h ; 计算量 $Q = 4.5 \text{ t/h}$; $P = 0.075 \text{ atm}$ (55 毫米汞柱)
数量	1
尺寸、滞留量	裕量系数 1.4

4. 大气冷凝器 混合硫酸盐结晶 · 位号 4.1.25

用途	冷凝二次蒸汽，采用冷却循环水喷淋
主要参数	流量 9.1 t/h ; 计算量 $Q = 12.0 \text{ t/h}$; $P = 0.075 \text{ atm}$
数量	1
尺寸、滞留量	裕量系数 1.3

5. 蒸汽喷射真空机组 (SJVV) 结晶 · 位号 6 , 图 4.4/4.5

型式/型号	喷射器 + 真空机组冷凝器 + 真空泵
用途	保证冷凝器内的真空
数量	每条生产线 1 台

报价中须确认：工况介质、温度、压力、接触部件材质、生产能力、外形尺寸及重量、能耗、制造周期、交货条件。